

Assignment 3

Mathematics for Cryptography

2025

For the best learning outcome and learning experience, use all resources from the course. Take notes on what you still need to learn by heart, what you should practice, and where you need to deepen your understanding. In your problem-solving, ensure that you understand each step and that there is a reason for taking that step in the given context. **Please provide explanations, not just computations, in your submission.**

We hope you enjoy this final assignment of the course and find it useful.

Problem 1

We prove that $\varphi(pq) = (p-1)(q-1)$ in three steps. Please follow the instructions below.

- A. Recall the definition of the Euler's totient function $\varphi(n)$ in your own words. Put this definition in the context of $n = pq$, where p and q are distinct prime numbers.
- B. Count the integers that are NOT coprime to pq . *Hint:* First, count the multiples of p . Then, count the multiples of q . Finally, count the multiples of both p and q .
- C. Based on the result in B., calculate $\varphi(pq)$ and simplify the expression to the required form.

Problem 2

- (a) To sign a message and then to verify the digital signature, follow the procedure of the RSA signature described on Slide 15 of Week 5 with the given parameters:

$$n = 851, \quad e = 17, \quad H = 240^1$$

Hint: You will need the private exponent d to sign the message, for which you need some prior steps.

¹To make this toy exercise somewhat realistic, we chose the message $m =$ "This is a message to be signed using the RSA signature." and used the cryptographic SHA-256 resulting in `db556275cabdb940a5938f2f2f214de56d8c424851ade3c714dbfdc4d74cfcf0`. (UTF-8 input encoding, hex lower case output). Since we can't compute with such large numbers in a manual problem, we took the very end of the hash output `f0`, which corresponds to 240 in decimal.

- (b) To make the exponentiation more efficient, the Chinese Remainder Theorem (CRT) is often used in practice. In the previous subproblem, you had to calculate an exponentiation to sign the hashed message. To use CRT, compute the exponentiation modulo both prime factors of n , p and q , and recombine them with Garner's Method (see page 231 in the Lecture Notes) to get signature s modulo n . Is this the same as in (a)?

Problem 3

As a reminder, we first give you the definition of an elliptic curve: Let $p \geq 5$ be a prime, then an elliptic curve in *Weierstrass representation* is of the form:

$$y^2 = x^3 + Ax + B \pmod{p}$$

where $A, B \in \mathbb{Z}_p$ satisfy $4A^3 + 27B^2 \pmod{p} \neq 0$. Let $E(\mathbb{Z}_p)$ be the set of points (x, y) satisfying the above equation and the point infinity ∞ , that is,

$$E(\mathbb{Z}_p) = \{(x, y) \mid x, y \in \mathbb{Z}_p \text{ and } y^2 = x^3 + Ax + B \pmod{p}\} \cup \{\infty\}$$

The elements in $E(\mathbb{Z}_p)$ are called points on the elliptic curve E .

Let $f(x) = x^3 + 2x + 4$ and consider the curve $E : y^2 = f(x) \pmod{7}$.

- (a) Check that the discriminant $4A^3 + 27B^2 \not\equiv 0 \pmod{p}$ for this curve.
 (b) Calculate all the points on the curve E .

Hint: start by calculating all possible squares mod 7 for all $x \in \mathbb{Z}_7$, after which you can check the right-hand side with various values of x .

- (c) One of the points on the curve is $(2, 3)$. Show that $\text{ord}((2, 3)) = 5$.

For your convenience, we provide Table 11.1 from the lecture notes below.

$P = -Q$	$P \neq \pm Q$	$P = Q$
∞	$\lambda = \frac{y_0 - y_1}{x_0 - x_1}$ $x_2 = \lambda^2 - x_0 - x_1$ $y_2 = -y_0 + \lambda(x_0 - x_2)$	$\lambda = \frac{3x_0^2 + A}{2y_0}$ $x_2 = \lambda^2 - 2x_0$ $y_2 = -y_0 + \lambda(x_0 - x_2)$

- (d) The Diffie–Hellman key agreement (DH) can be described in a prime group as follows:

Alice	Bob
share in advance:	p, g, q
$a \in_R \mathbb{Z}_q$ $A := g^a \pmod{p}$	$b \in_R \mathbb{Z}_q$ $B := g^b \pmod{p}$
$\xrightarrow{\text{Alice}, A}$ $\xleftarrow{\text{Bob}, B}$	
$K_{A,B} := B^a \pmod{p}$	$K_{B,A} := A^b \pmod{p}$

Give an analogous description of DH in an elliptic-curve cryptography (ECC) setting.

- (e) Assume that $(2, 3)$ is a base point, and Alice and Bob use 2 and 3 as secret values, respectively. Describe the elliptic-curve DH with these parameters.